ADITYA UNIVERSITY, SURAMPALEM

School of Computing

Department of Artificial Intelligence and Machine Learning

DATA ANALYSIS ESSENTIALS

FOOD DELIVERY DATA ANALYSIS SYSTEM

A Cornerstone / Group Project Documentation

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Academic Year: 2026–2027

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CERTIFICATE

This is to certify that the Cornerstone project work entitled "FOOD DELIVERY DATA ANALYSIS SYSTEM" is submitted by **B. BHANU PRAKASH REDDY (25B11AI102), M. GANESH REDDY (25B11AI772), T. SANTOSH REDDY (25B11AIC08), and G. ROHITH REDDY (25B11AI388)** in partial fulfilment of the requirements for the award of the B.Tech. degree in Artificial Intelligence and Machine Learning at Aditya University, Surampalem, for the academic year 2026–2027.

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DECLARATION

We hereby declare that the project entitled “FOOD DELIVERY DATA ANALYSIS SYSTEM” is our original academic work and is submitted to ADITYA UNIVERSITY, Surampalem, in partial fulfillment of the requirements for the award of the B.Tech. degree in Artificial Intelligence and Machine Learning. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree or diploma in this or any other institution. We confirm that the analysis, documentation, visualizations, and code presented in this report are prepared for academic purposes.

Place: Surampalem
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ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, Sophia, for her guidance, encouragement, and support throughout the development of the Food Delivery Data Analysis System. Her suggestions helped us understand the process of collecting, cleaning, analyzing, and presenting data effectively.

We also thank the Department of Artificial Intelligence and Machine Learning, School of Computing, Aditya University, for providing the academic environment and resources required to complete this Data Analysis Essentials project.

Finally, we thank our faculty members, classmates, friends, and family for their valuable support and encouragement during the preparation of the project and documentation.

ABSTRACT

The Food Delivery Data Analysis System is a Python-based data analysis project developed for the Data Analysis Essentials subject. The project focuses on analyzing food delivery records containing order details, restaurant names, order values, delivery times, customer ratings, order date/time, and locations. The main purpose is to convert raw food delivery data into meaningful insights about restaurant performance, delivery efficiency, and customer satisfaction.

The proposed workflow includes dataset loading, data cleaning, validation, exploratory data analysis, restaurant-wise order and sales analysis, delivery-time analysis, customer-rating analysis, visualization, and interpretation of results. Python libraries such as Pandas and NumPy can be used for data handling and numerical analysis, while Matplotlib and Seaborn can be used to create clear visualizations. The presentation also identifies delivery-time prediction using Linear Regression, Decision Tree, or Random Forest as a future enhancement.

The sample presentation values are illustrative and should be replaced with values calculated from the final dataset before submission. The project demonstrates how data analysis can support better understanding of orders, restaurant performance, delivery operations, and customer satisfaction.

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Chapter 1 Introduction

1.1 Introduction to the Project

Food delivery platforms generate large amounts of data every day. The project uses food delivery records to study orders, restaurants, order values, delivery times, customer ratings, order date/time, and location. The purpose is to transform raw records into understandable information through cleaning, analysis, visualization, and interpretation.

The project follows a simple data-analysis pipeline: Food Delivery Data → Data Cleaning → Data Analysis → Visualization → Insights → Better Decisions.

1.2 Problem Statement

Delivery applications record huge amounts of data such as orders, restaurants, sales, delivery times, and customer ratings. Manually checking this data is slow and error-prone, and useful patterns may remain hidden. The project therefore addresses the question:

"How can food delivery data be analyzed to identify restaurant performance, delivery efficiency, and customer satisfaction?"

1.3 Objectives of the Project

1. Analyze the overall order activity in the dataset.
2. Compare restaurant-wise order counts and sales.
3. Study average, minimum, maximum, and grouped delivery times.
4. Analyze customer ratings and their distribution.
5. Examine the relationship between delivery time and customer rating.
6. Present findings through clear and understandable charts.
7. Provide a foundation for future delivery-time prediction.

1.4 Scope of the Project

- Food delivery order and restaurant performance analysis.
- Data cleaning and validation before analysis.
- Exploratory analysis using numerical and categorical fields.
- Visualization of orders, sales, delivery times, and ratings.
- Identification of useful patterns for decision support.
- Future extension toward delivery-time prediction and real-time analytics.

Chapter 2 Dataset Description and Understanding

2.1 Dataset Source

The PPT describes the dataset as food delivery records with one row per order. The final external dataset source is not specified in the PPT; therefore, the documentation treats the supplied sample records as illustrative and recommends replacing them with the final project dataset before final submission.

2.2 Dataset Description

Each row represents one food delivery order. The dataset contains information needed to study order activity, restaurant performance, delivery efficiency, and customer satisfaction.

2.3 Features / Attributes

Feature	Description	Data Type (recommended)
Order ID	Unique identifier for an order	String
Restaurant Name	Name of the restaurant	Categorical
Order Value	Value of the order in ₹	Numeric
Delivery Time	Delivery duration in minutes	Numeric
Customer Rating	Customer rating for the order/service	Numeric
Order Date / Time	Date and time when the order was placed	Datetime
Location	Order/delivery location or sector	Categorical

2.4 Target Variable, if Applicable

The core DAE analysis is descriptive and does not require a single target variable. For the future delivery-time prediction enhancement, Delivery Time can be treated as the target variable, with features such as restaurant, location/distance, order value, order time, and past delivery patterns.

2.5 Dataset Statistics / Initial Exploration

The PPT provides illustrative sample values rather than final measured results. The sample presentation shows 1,200 total orders, an average order value of ₹465, an average delivery time of 36 minutes, and an average rating of 4.2/5. These values are explicitly marked as illustrative and should not be treated as final results.

Metric	Illustrative PPT Value	Status
Total Orders	1,200	Sample / illustrative
Average Order Value	₹465	Sample / illustrative
Average Delivery Time	36 min	Sample / illustrative
Average Rating	4.2 / 5	Sample / illustrative
Minimum Delivery Time	18 min	Sample / illustrative
Maximum Delivery Time	62 min	Sample / illustrative

Chapter 3 Data Preprocessing

Data preprocessing prepares raw records for reliable analysis. The PPT identifies loading, missing-value checking, duplicate removal, correction of inconsistent entries, data-type conversion, and outlier checking as the main preprocessing steps.

3.1 Data Collection / Loading

A CSV or similar tabular file can be loaded into a Pandas DataFrame. Pandas provides convenient functions for reading, inspecting, filtering, grouping, and transforming tabular data.

3.2 Handling Missing Values

- Identify missing values using DataFrame.isnull() or DataFrame.isna().
- Measure missing values column-wise.
- Remove records only when the missing information cannot reasonably be recovered.
- For suitable numeric fields, use an appropriate replacement strategy when justified by the dataset.

3.3 Handling Duplicate / Inconsistent Data

- Check duplicate rows using duplicated().
- Remove exact duplicate records when they represent repeated entries.
- Check spelling and naming consistency in restaurant and location fields.
- Check that ratings, order values, and delivery times fall within sensible ranges.

3.4 Data Type Conversion

Order Date / Time should be converted to a datetime type. Order Value, Delivery Time, and Customer Rating should be numeric so that mathematical calculations and visualizations can be performed correctly.

3.5 Outlier Checking

Delivery time and order value may contain unusually high or low observations. Box plots, summary statistics, and percentile-based checks can be used to identify potential outliers. Outliers should be reviewed rather than automatically deleted.

3.6 Other Project-Specific Preprocessing

- Create useful time fields such as hour or meal period from Order Date / Time.
- Group delivery times into ranges such as 15–25, 25–35, 35–45, 45–55, and 55–65 minutes when required for presentation.
- Create restaurant-wise aggregates such as order count, total sales, and average sales.
- Prepare clean numeric columns for correlation or relationship analysis.

Chapter 4 System Design and Methodology

4.1 Proposed Methodology

8. Load the food delivery dataset.
9. Inspect structure, columns, data types, and missing values.
10. Clean duplicates, inconsistent entries, and invalid data.
11. Perform exploratory data analysis.
12. Analyze order activity and restaurant-wise performance.
13. Analyze delivery-time distribution and summary statistics.
14. Analyze customer ratings and compare ratings with delivery time.
15. Create visualizations for important findings.
16. Interpret the results and identify useful insights.

4.2 System Workflow / Architecture

Stage	Main Activity	Output
1. Dataset	Load raw food delivery records	Raw DataFrame
2. Cleaning	Missing values, duplicates, types, validation	Clean DataFrame
3. EDA	Summary statistics and distributions	Exploration results
4. Order Analysis	Order counts and activity	Order insights
5. Restaurant Analysis	Orders and sales by restaurant	Restaurant insights
6. Delivery Analysis	Average, min/max and grouped times	Delivery insights
7. Rating Analysis	Average ratings and time-rating comparison	Customer insights
8. Visualization	Charts and plots	Visual insights
9. Reporting	Interpretation and conclusions	Final report

4.3 Flowchart

Raw Data → Load with Pandas → Clean & Validate → Explore with NumPy/Pandas → Analyze → Visualize with Matplotlib/Seaborn → Interpret → Report

4.4 Tools and Technologies Used

Tool / Technology	Purpose
Python	Main programming language for data analysis
Jupyter Notebook / Google Colab	Development and execution environment
CSV / tabular dataset	Input data format
Pandas	Data loading, cleaning, grouping, aggregation and analysis
NumPy	Numerical operations and statistical calculations
Matplotlib	Charts and basic visualizations
Seaborn	Statistical and analytical visualizations

4.5 Python Libraries Used

The PPT does not explicitly list library names, but the requested implementation is naturally supported by the following Python modules. They should be retained only if they are actually used in the final code.

Library	Typical Use in This Project

pandas	read_csv(), DataFrame operations, missing values, duplicates, groupby(), aggregation
numpy	Numerical calculations, arrays, mean/median and mathematical operations
matplotlib.pyplot	Bar charts, histograms, line charts, scatter plots
seaborn	Distribution plots, box plots, relationship plots and cleaner statistical charts

4.6 Analysis Methods

- Descriptive statistics: count, mean, minimum, maximum and distribution.
- Group-by analysis: restaurant-wise orders and sales.
- Time analysis: delivery-time ranges and order-time patterns.
- Rating analysis: average rating and rating distribution.
- Relationship analysis: delivery time versus average customer rating.
- Visualization: bar graph, histogram, pie chart, box plot, line chart and scatter plot as applicable.

Chapter 5 Implementation

5.1 Development Environment

The project can be implemented in Google Colab or Jupyter Notebook using Python. The environment should contain Pandas, NumPy, Matplotlib, and Seaborn. The final notebook should execute from top to bottom without manual data manipulation.

5.2 Dataset Loading and Exploration

The first implementation stage is to import the libraries and load the dataset. Basic commands such as head(), shape, columns, info(), describe(), and isnull().sum() help understand the dataset.

5.3 Preprocessing Implementation

The preprocessing stage checks missing values, duplicates, data types, inconsistent values, and possible outliers. Date/time fields are converted into datetime format and numeric fields are converted into numeric data types before calculations.

5.4 Exploratory Data Analysis

- Calculate total number of orders.
- Calculate total and average order value.
- Calculate average, minimum and maximum delivery time.
- Calculate average customer rating.
- Count orders by restaurant.
- Calculate total and average sales by restaurant.
- Inspect delivery-time and rating distributions.

5.5 Order and Restaurant Analysis

Restaurant-wise analysis identifies how many orders each restaurant receives and compares total or average sales. The PPT gives sample restaurant counts: Spice Villa 310, Pizza Point 265, Burger Hub 215, Biryani House 180, Cafe Aroma 140, and Noodle Bar 90. These are sample values only.

5.6 Delivery Time Analysis

The PPT illustrates an average delivery time of 36 minutes, a minimum of 18 minutes, and a maximum of 62 minutes. It groups delivery times into ranges and shows 35–45 minutes as the most common sample range. Final values must be generated from the actual dataset.

5.7 Customer Rating Analysis

The presentation illustrates an average rating of 4.2/5 and shows a possible pattern in which ratings tend to decrease as delivery time increases. This is presented as an illustrative relationship, not a confirmed causal conclusion.

5.8 Important Code Explanation

A representative implementation structure is provided in the Appendix. It demonstrates loading data with Pandas, numerical operations with NumPy, grouping and aggregation, and visualization with Matplotlib and Seaborn.

Chapter 6 Data Visualization and Results

6.1 Bar Graph

A bar graph can show the number of orders received by each restaurant. It makes it easy to compare restaurant activity and identify restaurants with higher order counts.

6.2 Histogram

A histogram can show the distribution of delivery times. The PPT groups delivery times into ranges to identify the most common delivery-time interval.

6.3 Pie Chart

A pie chart can be used to show the proportion of orders contributed by major restaurants when the number of categories is limited.

6.4 Box Plot

A box plot can summarize delivery-time or order-value distributions and help identify unusually high or low observations.

6.5 Line / Trend Chart

A line chart can be used when the analysis needs to display order activity or average delivery time across dates or time periods.

6.6 Scatter Plot

A scatter plot can compare delivery time with customer rating. The PPT uses this comparison to study whether faster delivery is associated with higher ratings.

6.7 Project-Specific Visualizations

- Restaurant-wise number of orders.
- Restaurant-wise sales.
- Delivery-time distribution.
- Delivery-time versus average rating.
- Summary KPI cards for total orders, average order value, average delivery time, and average rating.

6.8 Interpretation of Visualizations

Visualizations make numerical patterns easier to communicate. According to the PPT's illustrative sample, a few restaurants receive a large share of orders, delivery times vary substantially, and quicker delivery is associated with better ratings in the sample. These statements must be recalculated from the final dataset before being reported as actual project findings.

Chapter 7 Analysis of Results

7.1 Key Analysis Metrics

KPI	Illustrative Value in PPT	Meaning
Total Orders	1,200	Overall order activity
Average Order Value	₹465	Typical order value
Average Delivery Time	36 min	Typical delivery duration
Average Rating	4.2 / 5	Overall customer rating
Minimum Delivery Time	18 min	Fastest sample delivery
Maximum Delivery Time	62 min	Slowest sample delivery

7.2 Illustrative Results from PPT

The presentation explicitly labels these as sample/illustrative values. They are included here to preserve the project presentation content, not as verified final measurements.

7.3 Result Interpretation

- Restaurant order counts can reveal differences in demand across restaurants.
- Restaurant-wise sales can show which restaurants contribute more revenue.
- Delivery-time statistics show the spread and typical speed of service.
- Customer ratings provide an indicator of perceived service quality.
- Comparing delivery time with ratings can reveal a possible association that can be investigated further.

7.4 Discussion

The analysis provides a structured way to understand food delivery operations. However, descriptive relationships should not automatically be interpreted as causal relationships. For example, a relationship between longer delivery times and lower ratings could also be influenced by location, order time, restaurant workload, or other factors.

Chapter 8 Output Screenshots

Insert the actual screenshots from the final Google Colab/Jupyter execution in this chapter. The documentation template contains an output-screenshot chapter, and the project PPT presents the expected analysis stages.

8.1 Input / Dataset Screenshots

[INSERT 8.1 INPUT / DATASET SCREENSHOTS SCREENSHOT HERE]

Insert screenshot of the raw dataset loaded into the notebook.

8.2 Preprocessing Outputs

[INSERT 8.2 PREPROCESSING OUTPUTS SCREENSHOT HERE]

Insert screenshots showing missing-value checks, duplicate checks, cleaned data, and data types.

8.3 Visualization Outputs

[INSERT 8.3 VISUALIZATION OUTPUTS SCREENSHOT HERE]

Insert screenshots of restaurant order charts, delivery-time distribution, rating plots, and other final visualizations.

8.4 Analysis Outputs

[INSERT 8.4 ANALYSIS OUTPUTS SCREENSHOT HERE]

Insert screenshots showing final KPI values and analysis summaries.

Chapter 9 Conclusion and Future Enhancements

9.1 Conclusion

The Food Delivery Data Analysis System demonstrates how raw food delivery records can be cleaned, explored, analyzed, and visualized using Python. The project focuses on order activity, restaurant performance, delivery efficiency, and customer satisfaction. Pandas and NumPy support data preparation and numerical analysis, while Matplotlib and Seaborn support visualization. The resulting workflow converts raw data into clear insights that are easier to explain and use for decision-making.

9.2 Limitations

- The PPT uses illustrative/sample values; final measured results depend on the actual dataset.
- The PPT does not specify a confirmed external dataset source.
- Descriptive analysis does not prove causation.
- Delivery-time prediction is presented as a future enhancement rather than a completed core analysis.
- The quality of insights depends on the completeness and accuracy of the input data.

9.3 Future Enhancements

- Real-time analytics dashboard.
- Delivery-time prediction using Linear Regression, Decision Tree, or Random Forest.
- Restaurant recommendation system.
- Customer segmentation.
- Real-time delivery monitoring.
- Additional features such as distance, traffic, weather, restaurant workload, and historical delivery patterns where data is available.

Chapter 10 References

1. Food Delivery Data Analysis System – project presentation supplied for the Data Analysis Essentials subject.

2. Data Analysis Essentials – documentation structure supplied by the user.

3. Python, Pandas, NumPy, Matplotlib, and Seaborn documentation may be consulted for implementation details in the final notebook.

Chapter 11 Appendix – Source Code

The following is a compact representative implementation. Update the filename and column names to match the actual final dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("food_delivery_data.csv")

# Initial exploration
print(df.head())
print(df.info())
print(df.isnull().sum())

# Remove duplicate records
df = df.drop_duplicates()

# Convert data types
df["Order Value"] = pd.to_numeric(df["Order Value"], errors="coerce")
df["Delivery Time"] = pd.to_numeric(df["Delivery Time"], errors="coerce")
df["Customer Rating"] = pd.to_numeric(df["Customer Rating"], errors="coerce")
df["Order Date / Time"] = pd.to_datetime(
    df["Order Date / Time"], errors="coerce"
)

# Remove rows missing essential analysis fields
df = df.dropna(subset=[
    "Restaurant Name", "Order Value",
    "Delivery Time", "Customer Rating"
])

# Basic KPIs
total_orders = len(df)
avg_order_value = df["Order Value"].mean()
avg_delivery_time = df["Delivery Time"].mean()
avg_rating = df["Customer Rating"].mean()

print("Total Orders:", total_orders)
print("Average Order Value:", round(avg_order_value, 2))
print("Average Delivery Time:", round(avg_delivery_time, 2))
print("Average Rating:", round(avg_rating, 2))

# Restaurant-wise analysis
restaurant_orders = df["Restaurant Name"].value_counts()
restaurant_sales = df.groupby("Restaurant Name")["Order Value"].sum()

print(restaurant_orders)
print(restaurant_sales)

# Delivery-time analysis
print(df["Delivery Time"].describe())

# Restaurant order chart
```

```
restaurant_orders.plot(kind="bar", figsize=(9, 5))
plt.title("Restaurant-wise Number of Orders")
plt.xlabel("Restaurant")
plt.ylabel("Orders")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Delivery-time distribution
plt.figure(figsize=(8, 5))
sns.histplot(df["Delivery Time"], bins=10, kde=True)
plt.title("Delivery Time Distribution")
plt.xlabel("Delivery Time (minutes)")
plt.ylabel("Number of Orders")
plt.show()

# Delivery time vs rating
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=df,
    x="Delivery Time",
    y="Customer Rating"
)
plt.title("Delivery Time vs Customer Rating")
plt.xlabel("Delivery Time (minutes)")
plt.ylabel("Customer Rating")
plt.show()
```

Note: This appendix is a representative code structure based on the analysis workflow in the PPT. The final submission should include the exact code executed on the final dataset and the resulting screenshots.